



Chapter 3

Determinants & Diagonalization

MAE101 - ALGEBRA

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Definition

1 Definition and Properties

Given the $n \times n$ matrix A , we denote the $(n - 1) \times (n - 1)$ matrix $A(i | j)$ obtained from A by deleting row i and column j of A .

Example. For $A = \begin{bmatrix} 1 & 2 & 3 & 2 \\ 3 & 4 & 2 & 5 \\ 6 & 7 & 1 & 3 \\ 9 & 2 & 10 & 4 \end{bmatrix}$. Find matrices $A(1 | 2)$ and $A(2 | 3)$?

Definition

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Solution.

$$A(1 | 2) = \begin{bmatrix} 3 & 2 & 5 \\ 6 & 1 & 3 \\ 9 & 10 & 4 \end{bmatrix}; \quad A(2 | 3) = \begin{bmatrix} 1 & 2 & 2 \\ 6 & 7 & 3 \\ 9 & 2 & 4 \end{bmatrix}$$

Determinant (Định thức)

1 Definition and Properties

For $A \in M_n(\mathbb{R})$. The **determinant** of matrix A , denoted as **$\det A$** or **$|A|$** is a real number.

- If $n = 1$, $A = (a)$, then **$\det A = a$** .
- If $n = 2$, $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then **$\det A = ad - bc$** .

- If $n \geq 3$, $A = \begin{pmatrix} a_{1,1} & a_{1,2} & \cdots & a_{1,n} \\ a_{2,1} & a_{2,2} & \cdots & a_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n,1} & a_{n,2} & \cdots & a_{n,n} \end{pmatrix}$, then

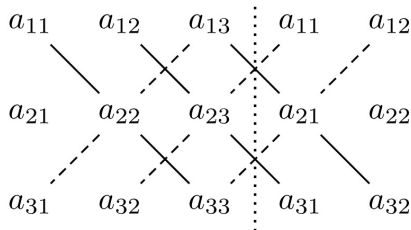
$$\det A \stackrel{\text{row } i}{=} \sum_{k=1}^n a_{ik}(-1)^{i+k}|A(i \mid k)| \stackrel{\text{column } j}{=} \sum_{k=1}^n a_{kj}(-1)^{k+j}|A(k \mid j)|$$

Determinant - Sarrus Rule (Only $n=3$)

1 Definition and Properties

If $n = 3$, $A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$, then

$$\det A = [a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}] - [a_{31}a_{22}a_{13} + a_{32}a_{23}a_{11} + a_{33}a_{21}a_{12}]$$



Determinant - Example

1 Definition and Properties

Calculate the determinant of the matrix:

$$1. A = \begin{bmatrix} 4 & -2 \\ 3 & 5 \end{bmatrix}$$

$$2. B = \begin{bmatrix} 3 & -1 & 3 \\ 5 & 2 & 2 \\ 4 & 1 & 0 \end{bmatrix}$$

$$3. C = \begin{bmatrix} 2 & -3 & 3 & 2 \\ 3 & 0 & 1 & 4 \\ -2 & 0 & 0 & 2 \\ 4 & 0 & -1 & 5 \end{bmatrix}$$

Determinant - Proposition

1 Definition and Properties

Let $A \in M_n(\mathbb{R})$.

1. $\det A^T = \det A$.
2. If A has a row or column of 0 , then $\det A = 0$.
3. If A is a square triangular matrix, then $\det A$ is the product of the entries on the main diagonal. That is, $\det A = a_{11} \cdot a_{22} \dots a_{nn}$

Example. Calculate the determinant of the following matrices:

$$A = \begin{bmatrix} -1 & 0 & 4 \\ 3 & 0 & 4 \\ 4 & 0 & -2 \end{bmatrix}; \quad B = \begin{bmatrix} 2 & 3 & 4 \\ 0 & -3 & 9 \\ 0 & 0 & 4 \end{bmatrix}; \quad C = \begin{bmatrix} -2 & 0 & 0 \\ 3 & 3 & 0 \\ 9 & 8 & -5 \end{bmatrix}$$

Determinant - Proposition

1 Definition and Properties

Let $A \in M_n(\mathbb{R})$.

1. $\det A^T = \det A$.
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Example. Calculate the determinant of the following matrices:

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Solution.

$$\det A = 0; \quad \det B = 2 \cdot (-3) \cdot 4 = -24; \quad \det C = (-2) \cdot 3 \cdot (-5) = 30.$$



Theorem

1 Definition and Properties

Theorem

Given $A, A' \in M_n(\mathbb{R})$. Then,

1. If $A \xrightarrow{r_i \leftrightarrow r_j} A'$, then $|A'| = -|A| \iff |A| = -|A'|$, for any $i \neq j$,
2. If $A \xrightarrow{\alpha r_i} A'$ then $|A'| = \alpha|A| \iff |A| = \frac{1}{\alpha}|A'|$,
3. If $A \xrightarrow{r_i + \beta r_j} A'$, then $|A'| = |A|$, for any $i \neq j$.

Example. If $\begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = 6$, evaluate $\begin{vmatrix} a+x & b+y & c+z \\ 3x & 3y & 3z \\ -p & -q & -r \end{vmatrix}$.

Solution

1 Definition and Properties

$$\begin{aligned}
 & \begin{vmatrix} a+x & b+y & c+z \\ 3x & 3y & 3z \\ -p & -q & -r \end{vmatrix} \xleftrightarrow{d_2 \leftrightarrow d_3} \begin{vmatrix} a+x & b+y & c+z \\ -p & -q & -r \\ 3x & 3y & 3z \end{vmatrix} \\
 & \xleftrightarrow{d_2 =: -d_2} \begin{vmatrix} a+x & b+y & c+z \\ p & q & r \\ 3x & 3y & 3z \end{vmatrix} \xleftrightarrow{d_3 =: \frac{1}{3}d_3} \begin{vmatrix} a+x & b+y & c+z \\ p & q & r \\ x & y & z \end{vmatrix} \\
 & \xleftrightarrow{d_1 =: d_1 - d_3} \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix} = 3 \cdot 6 = 18.
 \end{aligned}$$



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The cofactor (phần bù đại số)

2 Determinants and Invertible Matrix

Given the $n \times n$ matrix A . For each $i, j \in \overline{1, n}$, we call $c_{ij} = (-1)^{i+j} \det A(i | j)$ is **the cofactor** of coefficient a_{ij} .

Example. For $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 3 & 4 & 0 \end{bmatrix}$. Find the cofactors of a_{12} and a_{31} ?

The cofactor (phần bù đại số)

2 Determinants and Invertible Matrix

Given the $n \times n$ matrix A . For each $i, j \in \overline{1, n}$, we call $c_{ij} = (-1)^{i+j} \det A(i | j)$ is **the cofactor** of coefficient a_{ij} .

Example. For $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 3 & 4 & 0 \end{bmatrix}$. Find the cofactors of a_{12} and a_{31} ?

Solution.

$$c_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 1 \\ 3 & 0 \end{vmatrix} = 3, \quad c_{31} = (-1)^{3+1} \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} = -2$$

Adjugate matrix (ma trận phụ hợp)

2 Determinants and Invertible Matrix

For $A = (a_{ij}) \in M_n(\mathbb{R})$. Put $C = (c_{ij})$ with c_{ij} is the cofactor of a_{ij} . We call the transpose matrix C^T of C is **adjugate matrix** of A , denoted as **adj**(A).

Example. For $A = \begin{bmatrix} 2 & 3 & 1 \\ 2 & -1 & 2 \\ 3 & 4 & -2 \end{bmatrix}$. Find the adjugate matrix of A .

Adjugate matrix (ma trận phụ hợp)

2 Determinants and Invertible Matrix

For $A = (a_{ij}) \in M_n(\mathbb{R})$. Put $C = (c_{ij})$ with c_{ij} is the cofactor of a_{ij} . We call the transpose matrix C^T of C is **adjugate matrix** of A , denoted as **adj**(A).

Example. For $A = \begin{bmatrix} 2 & 3 & 1 \\ 2 & -1 & 2 \\ 3 & 4 & -2 \end{bmatrix}$. Find the adjugate matrix of A .

Solution. We have

$$C = \begin{bmatrix} -6 & 10 & 11 \\ 10 & -7 & 1 \\ 7 & -2 & -8 \end{bmatrix} \implies \text{adj}(A) = C^T = \begin{bmatrix} -6 & 10 & 7 \\ 10 & -7 & 1 \\ 11 & 1 & -8 \end{bmatrix}$$

Invertible Matrix

2 Determinants and Invertible Matrix

Theorem

1. If $A, B \in M_n(\mathbb{R})$, then $\det AB = \det A \det B$.
2. An $n \times n$ matrix A is invertible if and only if $\det A \neq 0$. When this is the case,

$$\det(A^{-1}) = \frac{1}{\det A}$$

Furthermore,

$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

Example. For $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 3 & 4 & 0 \end{bmatrix}$. Is matrix A invertible or not? If it is invertible, find the the invertible matrix of A .

Invertible Matrix - Solution

2 Determinants and Invertible Matrix

We have $\det A = -2 \neq 0$. Hence, A is invertible.

$$c_{11} = (-1)^{1+1} \begin{vmatrix} 3 & 1 \\ 4 & 0 \end{vmatrix} = -4, c_{12} = (-1)^{1+2} \begin{vmatrix} 2 & 1 \\ 3 & 0 \end{vmatrix} = 3, c_{13} = (-1)^{1+3} \begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = -1$$

$$c_{21} = (-1)^{2+1} \begin{vmatrix} 1 & 1 \\ 4 & 0 \end{vmatrix} = 4, c_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 1 \\ 3 & 0 \end{vmatrix} = -3, c_{23} = (-1)^{2+3} \begin{vmatrix} 1 & 1 \\ 3 & 4 \end{vmatrix} = -1$$

$$c_{31} = (-1)^{3+1} \begin{vmatrix} 1 & 1 \\ 3 & 1 \end{vmatrix} = -2, c_{32} = (-1)^{3+2} \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} = 1, c_{33} = (-1)^{3+3} \begin{vmatrix} 1 & 1 \\ 2 & 3 \end{vmatrix} = 1$$

$$C = \begin{bmatrix} -4 & 3 & -1 \\ 4 & -3 & -1 \\ -2 & 1 & 1 \end{bmatrix} \implies \text{adj}(A) = C^T = \begin{bmatrix} -4 & 4 & -2 \\ 3 & -3 & 1 \\ -1 & -1 & 1 \end{bmatrix}$$

$$\implies A^{-1} = \frac{1}{-2} \begin{bmatrix} -4 & 4 & -2 \\ 3 & -3 & 1 \\ -1 & -1 & 1 \end{bmatrix}$$

Example

2 Determinants and Invertible Matrix

Find all the values of m such that the following matrices are invertible

$$A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 2 & 1 & 5 & 3 \\ 5 & 0 & 7 & m \\ -1 & 2 & 3 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & m \\ 3 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 5 & 7 & 5 \end{bmatrix}$$

Example

2 Determinants and Invertible Matrix

Find all the values of m such that the following matrices are invertible

$$A = \begin{bmatrix} 1 & 1 & 2 & 1 \\ 2 & 1 & 5 & 3 \\ 5 & 0 & 7 & m \\ -1 & 2 & 3 & -3 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 3 & m \\ 3 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 2 & 3 & 2 \\ 5 & 7 & 5 \end{bmatrix}$$

Solution. We have

$\det(A) = 8m - 72$. Thus, the matrix A is invertible when

$$8m - 72 \neq 0 \iff m \neq 9.$$

$\det(B) = (4m - 4) \times 0 = 0$. Thus, the matrix B is not invertible for any m .



Proposition

2 Determinants and Invertible Matrix

Proposition

For $A, B \in M_n(\mathbb{R})$ and A is invertible.

1. $|A^{-1}| = \frac{1}{|A|},$

2. $|\alpha A| = \alpha^n |A|,$

3. $|\text{adj}(A)| = |A|^{n-1},$

4. $|AB| = |A||B|$

Example.

For $A, B \in M_3(\mathbb{R})$ and $|A| = 3, |B| = -2$. Calculate $|(2AB)^{-1}|$ and $|\text{adj}(AB)|$

Proposition

2 Determinants and Invertible Matrix

Proposition

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$$1. \quad |A^{-1}| = \frac{1}{|A|},$$

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Example.

For $A, B \in M_3(\mathbb{R})$ and $|A| = 3, |B| = -2$. Calculate $|(2AB)^{-1}|$ and $|\text{adj}(AB)|$

Solution.

$$|(2AB)^{-1}| = \frac{1}{|(2AB)|} = \frac{1}{2^3 |AB|} = \frac{1}{8 |A||B|} = \frac{1}{8 \times 3 \times (-2)} = -\frac{1}{48}$$

$$|\text{adj}(AB)| = |AB|^{3-1} = (|A||B|)^2 = (3 \times (-2))^2 = 36.$$



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Cramer's Rule

3 Cramer's Rule

Cramer's Rule

If A is an invertible $n \times n$ matrix, the solution to the system

$$A\mathbf{x} = \mathbf{b}$$

of n equations in the variables $x_1, x_2, \dots, x_n$ is given by

$$x_1 = \frac{\det A_1}{\det A}, x_2 = \frac{\det A_2}{\det A}, \dots, x_n = \frac{\det A_n}{\det A}$$

where, for each k , A_k is the matrix obtained from A by replacing column k by $\mathbf{b}$.

Example. Find x_1 , given the following system of equations.

$$\begin{cases} 5x_1 + x_2 - x_3 &= 4 \\ 9x_1 + x_2 - x_3 &= 1 \\ x_1 - x_2 + 5x_3 &= 2 \end{cases}$$

Cramer's Rule - Solution

3 Cramer's Rule

Compute the determinants of the coefficient matrix A and the matrix A_1 obtained from it by replacing the first column by the column of constants.

$$\det A = \det \begin{bmatrix} 5 & 1 & -1 \\ 9 & 1 & -1 \\ 1 & -1 & 5 \end{bmatrix} = -16$$
$$\det A_1 = \det \begin{bmatrix} 4 & 1 & -1 \\ 1 & 1 & -1 \\ 2 & -1 & 5 \end{bmatrix} = 12$$

Hence, $x_1 = \frac{\det A_1}{\det A} = -\frac{3}{4}$ by Cramer's rule.



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Eigenvalues & Eigenvectors

4 Eigenvalues and Eigenvectors

Let A is an $n \times n$ matrix

1. A number λ is called an **eigenvalue** of A if $\exists x \in \mathbb{R}^n \setminus \{0\} : Ax = \lambda x$.
2. x is called an **eigenvector** of A corresponding to the eigenvalue λ .
3. The **characteristic polynomial** $p_A(\lambda)$ of A is defined by

$$p_A(\lambda) = \det(\lambda I - A)$$

Theorem

1. The **eigenvalues** λ of A are the **roots** of the characteristic polynomial $p_A(\lambda)$ of A .
2. The λ -eigenvectors x are the nonzero solutions to the homogeneous system

$$(\lambda I - A)x = 0$$

of linear equations with $\lambda I - A$ as coefficient matrix.



Eigenvalues & Eigenvectors - Example

4 Eigenvalues and Eigenvectors

Find the $p_A(\lambda)$, eigenvalues and basic eigenvector for $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$.

Eigenvalues & Eigenvectors - Example

4 Eigenvalues and Eigenvectors

Find the $p_A(\lambda)$, eigenvalues and basic eigenvector for $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$.

Solution.

$$1. \ p_A(\lambda) = \det [\lambda I_2 - A] = \det \begin{bmatrix} \lambda + 1 & -3 \\ 2 & \lambda - 4 \end{bmatrix} = \lambda^2 - 3\lambda + 2$$

2. The eigenvalues are $\lambda_1 = 1$ and $\lambda_2 = 2$.

3. Find eigenvectors

• $\lambda_1 = 1$, the eigenvectors $x_1 = (u, v)$ are the solutions of the equation system:

$$(\lambda_1 I - A) x_1 = 0 \Leftrightarrow \begin{bmatrix} 2 & -3 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = 0 \Leftrightarrow \begin{cases} u = \frac{3}{2}t \\ v = t \end{cases} \Rightarrow x_1 = \left(\frac{3}{2}, 1\right)t, t \neq 0.$$

• $\lambda_2 = 2$, the eigenvectors $x_2 = (u, v)$ are the solutions of the equation system:

$$(\lambda_2 I - A) x_2 = 0 \Leftrightarrow \begin{bmatrix} 3 & -3 \\ 2 & -2 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = 0 \Leftrightarrow \begin{cases} u = t \\ v = t \end{cases} \Rightarrow x_2 = (1, 1)t, t \neq 0.$$



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Diagonalization

5 Diagonalization

An $n \times n$ matrix A is **diagonalizable** (chéo hóa được) if there is an invertible $n \times n$ matrix P such that

$$A = PDP^{-1}$$

Theorem

1. A is diagonalizable if and only if it has eigenvectors $x_1, x_2, \dots, x_n$ such that the matrix $P = \begin{bmatrix} x_1 & x_2 & \dots & x_n \end{bmatrix}$ is invertible.

2. When this is the case, $P^{-1}AP = D$ where $D = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}$ for each i, λ_i is the eigenvalue of A corresponding to x_i .

Diagonalization Algorithm

5 Diagonalization

To diagonalize an $n \times n$ matrix A :

- **Step 1.** Find the distinct **eigenvalues** λ of A .
- **Step 2.** Compute the **basic eigenvectors** corresponding to each of these eigenvalues λ as basic solutions of the homogeneous system $(\lambda I - A)x = 0$.
- **Step 3.** The matrix A is diagonalizable if and only if there are n basic eigenvectors in all.
- **Step 4.** If A is diagonalizable, then $n \times n$ matrix P with these basic eigenvectors as its columns is a diagonalizing matrix for A , that is, P is invertible and $P^{-1}AP$ is diagonal.

Computing High Powers of a Matrix

5 Diagonalization

Let A be an $n \times n$ matrix that is diagonalizable. That is, we can find an invertible matrix P such that $A = PDP^{-1}$ or, what is the same, $P^{-1}AP = D$ where D is diagonal. Suppose we want to compute A^{100} . By first diagonalizing A , we can compute A^{100} as follows:

$$\begin{aligned}
 A^{100} &= \left(PDP^{-1}\right)^{100} \\
 &= \underbrace{\left(PDP^{-1}\right) \left(PDP^{-1}\right) \left(PDP^{-1}\right) \cdots \left(PDP^{-1}\right)}_{100 \text{ times}} \\
 &= PD \left(P^{-1}P\right) D \left(P^{-1}P\right) D \left(P^{-1} \cdots P\right) DP^{-1} \\
 &= PD^{100}P^{-1}.
 \end{aligned}$$



Diagonalization - Example

5 Diagonalization

Diagonalize the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ in the late example and find A^{20} .

Diagonalization - Example

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Diagonalize the matrix $A = \begin{bmatrix} -1 & 3 \\ -2 & 4 \end{bmatrix}$ in the late example and find A^{20} .

Solution.

- Step 1. The eigenvalues of A are $\lambda_1 = 1$ and $\lambda_2 = 2$.
- Step 2. The corresponding basic eigenvectors are $x_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$, $x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- Step 3. The matrix $P = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$ and P is invertible.
- Step 4. The previous Theorem guarantees that $D = P^{-1}AP = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

$$A^{20} = PD^{20}P^{-1} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}^{20} \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} -2097149 & 3145725 \\ -2097150 & 3145726 \end{bmatrix}$$



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Problems

6 Problems

1. Evaluate the determinant

$$\text{a. } A = \begin{bmatrix} -2 & 0 & 0 \\ 4 & 6 & 0 \\ -3 & 7 & 2 \end{bmatrix} \quad \text{b. } B = \begin{bmatrix} -3 & 2 & 1 \\ 4 & 5 & 6 \\ 2 & -3 & 1 \end{bmatrix} \quad \text{c. } C = \begin{bmatrix} m & -1 & 0 \\ 1 & 2 & 1 \\ 2 & m & -3 \end{bmatrix}$$

2. Find the cofactors of the matrix $A = \begin{bmatrix} -3 & 4 & 2 \\ 6 & 3 & 1 \\ 4 & -7 & -8 \end{bmatrix}$.

3. Find the adjugate and the inverse of the matrix $A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \\ 2 & 1 & 0 \end{bmatrix}$.

4. Let $A = \begin{bmatrix} 1 & * & * & * \\ 0 & -1 & * & * \\ 0 & 0 & 2 & * \\ 0 & 0 & 0 & 2 \end{bmatrix}$. Find $|2A^{-1}|$, $|AA^T|$, $|\text{adj } A|$, $|-A^3|$, $|A^{-1} - 2 \text{adj } A|$.



Problems

6 Problems

5. Let A and B be square matrices of order 5 such that $|A| = -5$ and $|B| = 3$. Find

a. $|2AB|$ b. $|\text{adj}(AB)|$ c. $|5A^{-1}B^T|$ d. $|A^T B^{-1} A^2|$

6. Find all values of m for which the matrix is not invertible

a. $A = \begin{bmatrix} 1 & 3 \\ m & 2 \end{bmatrix}$ b. $B = \begin{bmatrix} m & 1 & 3 \\ 1 & 3 & 2 \\ -1 & 4 & 5 \end{bmatrix}$

7. Let $A = \begin{bmatrix} 1 & 3 & 1 \\ 0 & 1 & 0 \\ 2 & -1 & x \end{bmatrix}$. For which values of x is A invertible?

8. Find the eigenvalues, corresponding eigenvectors, and diagonalization of the matrices

a. $A = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$ b. $B = \begin{bmatrix} -3 & 2 & -1 \\ 0 & 1 & 0 \\ 4 & 1 & 1 \end{bmatrix}$



Q&A

Thank you for listening!